
Managing Sustainable Tourism: A Tourism Optimization Problem Based on Juneau

‘Every coin has its two sides’, and this also applies to the travel industry. As global tourism surges as the epidemic subsides, cities like Juneau, Alaska benefit economically but face environmental and social challenges. To address the conflicts between benefits and hidden costs, a sustainable tourism optimization model (**STM**) is proposed in the paper.

For Problem 1, four evaluation indicators are selected: **tourism revenue (benefit) and carbon emissions, infrastructure pressure, resident satisfaction (costs)**. The ideal tourist volume V_{opt} is also introduced as a reference standard. Each aspect is related to tourist volume V , which is taken as the independent variable, and the functional relationship between the four indicator functions and V is constructed and a series of related coefficients is introduced. By setting the four weights, they are weighted and summed up to get the **final objective Z** (multi-objective model transformed into a single objective model). By setting up a series of constraints and preconditions, the parameters in the equation are constantly changed so that Z can be maximized. The **Simulated Annealing Algorithm (SAA)** produces the result that the coefficient of tourism revenue is 606.7, requiring improved tourist spending; the coefficient of carbon emission is 0.18, which indicates a need for stronger environmental efforts.

For Question 2, the paper analyzes two cities: Barcelona, Spain (over-tourism) and Jian Shui, Yunnan, China (under-tourism). By applying the **STM** model with adjusted weights, it concludes that Barcelona should enhance its infrastructure and tourist capacity, while Jian Shui should focus on promotion to attract more tourists and improve infrastructure in preparation.

Next, the model undergoes **sensitivity analysis**, focusing on the impact of tourism revenue functions and changes in resident satisfaction coefficients. Two cases are explored: **logarithmic** (used in Problem 1) and **linear** tourism revenue functions. By adjusting the coefficients of f (related to satisfaction degree) across 10 parameter magnitudes, the sensitivity of the model to these changes is analyzed. Data visualization, including function graphs and optimization line charts, reveals patterns and informs optimization conclusions. Using the **ARIMA time series model** for forecasts, it predicts tourist numbers will rise to 1,740,000 in 2024 before declining in 2025.

To conclude, we propose the **STM** model to optimize sustainable tourism, using Juneau as a case study and applying it to other cities. Sensitivity analysis verifies key parameter impacts, and suggestions are given based on predictions of tourist volume.

Keywords: Sustainable development; Optimization of the tourism sector; Multi-objective optimization; Simulated annealing; ARIMA prediction

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MEMO

Date: 2025/1/26

Topic: Our Suggestions for Sustainable Juneau Tourism

Since the pandemic's recovery, Juneau, like many other tourist cities around the world, has faced new opportunities and challenges. In 2023, the number of tourists (1.67 million) has recovered to the level before the epidemic (or even higher), bringing a lot of economic benefits to the local area and driving local employment. However, there are good and bad sides to things. Hidden costs like increasing carbon emissions, growing infrastructure pressure, resident dissatisfaction can't be ignored. So we establish a tourism optimization model considering the underlined four parts to get the balance.

From the study, the overall benefits of Juneau's tourism are largely dependent on maintaining a high number of visitors. However, to ensure long-term sustainability, it's imperative to adopt a forward-looking approach that protects the natural landscapes. In a long-term perspective, the most crucial part should be the problems caused by environmental pollution and excessive carbon emissions, because these will lead to the disappearance of natural scenery such as glaciers, which will undoubtedly lead to Juneau losing an attractive place to play.

'Protect the ecological environment, not only for the earth's environment but also for our own.'

What to do next?

- a) **Increase Hotel Tax & Tourist Fees:** Enhance government revenue to provide financial support for environmental protection and infrastructure improvement.
- b) **Strengthen Environmental Protections:** Protect natural landmarks, advocate for low-carbon tourism, and promote eco-friendly transportation and accommodations.
- c) **Promote Lesser-Visited Attractions:** Redirect tourist flows to less-exploited sites through marketing to balance visitor distribution.

What will happen in the future?

Tourism volume in Juneau is projected to keep growing. We predict the volume in 2024 is 1742053, but a decline in 2025 (10978020) and 2026 (1475914). But don't worry we believe that the growing trend will continue and indicate continued economic benefits. However, the government and tourism commission must also prepare for potential disruptions (e.g., pandemics) to minimize future risks.

1 Introduction

1.1 Problem Background

In recent years, with the rapid economic recovery after the epidemic, the tourism industry will re-enter a phase of rapid development^[1]. Although the sharp increase in the number of tourists can enhance the economic income of a region and provide more local jobs, it cannot be ignored that this phenomenon brings about unbearable hidden costs. When the number of tourists exceeds the region's carrying capacity, the negative impacts will exceed the economic gains and job increases, hindering regional tourism development.



Figure 1. The greenhouse effect of melting glaciers makes animals homeless

The surge in tourists has significant impacts, which can be categorized into environmental and social impacts. The increase in tourist arrivals will lead to an increase in carbon emissions, the melting of glaciers, and the disappearance of famous tourist attractions (Figure 1). At the same time, the increase in tourists can lead to a sudden increase in pressure on infrastructure in areas with low levels of economic development, and the arrival of tourists can lead to protests by some of the inhabitants of these tourist towns, as their relaxed and comfortable environments are disrupted and spaces become crowded.

1.2 Restatement of the Problem

Juneau, Alaska, is the case we study to explore the challenges posed by the issues outlined above. As a tourist city grappling with the negative impacts of oversaturation in the tourism industry, Juneau faces the critical task of identifying effective measures and implementing policies that maximize the benefits of tourism while minimizing its hidden costs.

- **Problem 1:** A sustainable tourism model is first constructed around Juneau. Some factors related to the local tourism industry, including the number of tourist visitors and the revenue of the tourism industry are taken into consideration, and some constraints are proposed to make some of these factors optimized. The goal is to raise the revenue of tourism industry to the maximum with the minimum hidden cost. At the same time, a sensitivity analysis of the model is carried out to study the extent to which the results are affected by changes in the coefficients.
- **Problem 2:** Select another case that the tourism industry is "too prosperous", analyze it with the model in question 1, and propose what factors should be paid attention to in order to reduce the hidden cost. At the same time, for areas with fewer tourists, what measures should be taken to improve tourism industry income. The ultimate goal, of course, is to strike a balance between costs and benefits.

- **Problem 3:** Based on the tourism industry optimization model given previously, this paper will combine the past development of Juneau's tourism industry, forecast and analyze the performance of the tourism industry based on the optimization model, and put forward optimization suggestions for tourist council of Juneau.

1.3 Our Tasks

- ✓ **Task 1:** According to the current situation of Juneau's tourism industry, it is in a state of 'oversaturation'. In this paper, the tourism industry optimization model sets two objectives: tourism revenue & hidden cost. Hidden costs include the impact of carbon emissions, pressure on infrastructure, and residents' satisfaction.
- ✓ **Task 2:** Discuss the sensitivity of the model, adjust the model parameters, and study the law.
- ✓ **Task 3:** Different cases of tourism cities discuss different situations. In the case of Juneau, cities are often affected by the problem of 'oversaturation', and it is necessary to reduce hidden costs as much as possible while ensuring tourism industry income. For a tourist city with a small number of tourists, how to improve the income of the tourism industry (attract others to travel) is the focus of the model. In essence, regardless of the type of tourist city, the optimization model remains roughly the same.
- ✓ **Task 4:** Back to the example of Juneau, this paper will make a prediction for the future based on the past local tourism development situation and information. At the same time, according to the current development situation, a memo will be written for the Juneau Tourism Committee to tell them about the relevant research and prediction in this paper and give some valuable suggestions.

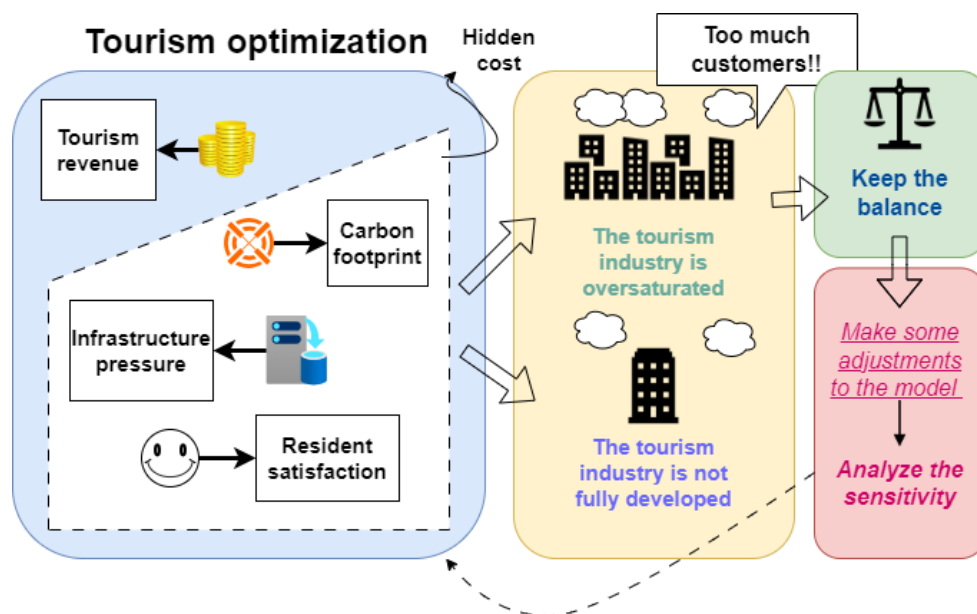


Figure 2. Task flow chart

2 Model Assumptions

In order to facilitate the establishment and quantitative calculation of the model, the following assumptions are set in this paper:

- In the tourism industry optimization model, the relevant variables involved in tourism income and hidden cost (carbon emission impact, infrastructure pressure, residents' satisfaction) are

- not independent of each other, and are related to the number of tourists.
- In the following, weights are set for the four indicators of the model in order to make the model more refined. In this paper, we do not consider using objective mathematical weight solving methods, but we will discuss the changes of the optimization model under different subjective weights.

3 Notation and Variables

Some indicators are not listed and will be explained in detail:

Table 1. Notation and its meaning

Notation	Meaning
$V/V(t)$	Passenger flow volume (in year t)
C_{avg}	Average tourist consumption
E_{avg}	Carbon emissions per tourist
P_1	Infrastructure pressure coefficient
P_0	Infrastructure pressure on local resident
f	Coefficient of satisfaction degree influence
V_{opt}	The most appropriate number of visitors

4 Optimization Model of Sustainable Tourism

Based on the above content, a specific **sustainable tourism industry optimization model (STM)** will be presented in this part, including evaluation indicators for the tourism industry, and calculation of industry benefits and costs. In the model combined with the actual situation, set a number of parameters affecting the total benefit, select the factors worth optimizing, and then get the best benefit.

4.1 Optimization Goals of the Tourism Industry

The tourism industry optimization model has two main objectives: tourism revenue (orange) & hidden cost (blue), and the ultimate goal is to maximize the benefits of the tourism industry. The benefits of the tourism industry are measured, including 4 important evaluation indicators (mentioned in 1.3), and the reasons for choosing them are explained in detail here:

Table 2. 4 Parts of STM Models

Index	Description
Tourism revenue	It's the most direct indicator of the tourism industry's contribution to the local economy. Juneau City with glacier scenery, rain forest, and other special tourism programs to attract tourists. In 2023, the consumption of passengers reached 375 million US dollars, creating considerable income for the local economy, vigorously promoting local economic development, and directly reflecting the economic value of the tourism industry.
Carbon footprint	The development of tourism has an impact on the environment. <u>A large number of tourists arrive in Juneau by boat, which consumes a lot of fuel and emits greenhouse gases.</u> In addition, tourists participating in various tourism activities in the local area, such as taking transportation to scenic spots, scenic facilities operation,

	etc., will also produce carbon emissions. These carbon emissions not only affect the local ecological environment but also affect global climate change, affect the sustainability of Juneau's tourism resources.
Infrastructure pressure	An influx of tourists can put pressure on local infrastructure, <u>which is directly related to the tourist experience and the proper functioning of the city.</u> Juneau in the tourist season, the road around the port is congested, the public transportation capacity is insufficient, the sewage treatment and garbage disposal pressure increases, and the public facilities in the scenic spot are also facing greater pressure. Excessive pressure on infrastructure will reduce tourists' satisfaction and hinder tourism development.
Resident satisfaction	<u>Local residents are important stakeholders in the development of the tourism industry.</u> Tourism has created employment opportunities for residents, increased residents' incomes, and improved the living standards of some residents. However, it also brings traffic congestion, environmental pollution, and other problems, affecting the quality of life of some residents. Residents' attitude towards and satisfaction with tourism will affect their support for the development of the tourism industry, which in turn will affect the sustainable development and long-term benefits of the tourism industry.

4.2 Optimization Models

In this section, the details of the optimization model will be given. The essence of the tourism industry optimization model (*STM*) is to focus on the above four parts and to obtain an optimal final result after weighing the interests of all parties. Before explaining this, here is the calculation method for the four components respectively. It should be noted that the discussion of tourist volume in this part is to put forward some opinions on the future development of the local tourism industry, so the analysis is carried out around a specific period (time in 2023), so it does not involve time changes and $V(t)$

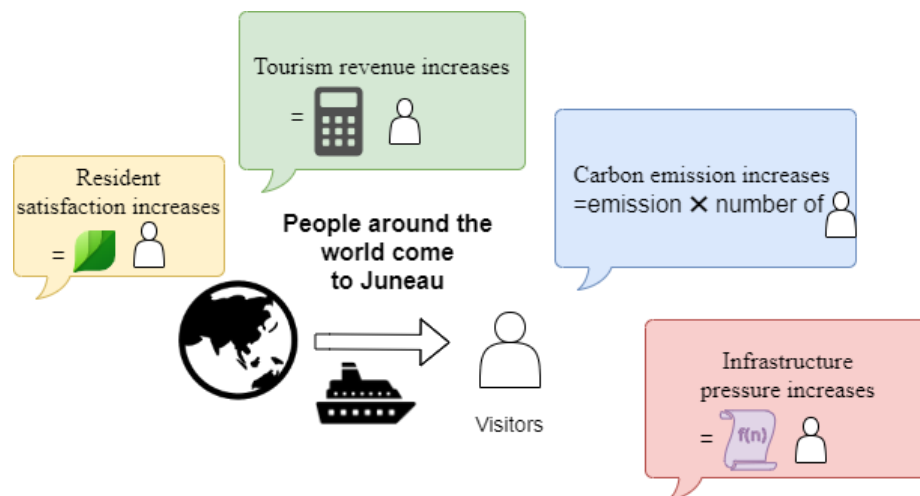


Figure 3. How to calculate each part

The four parts in this paper are all related to the tourists volume, so the four parts here can be constructed as the **correlation function of V** related to the number of tourists (Figure 3).

Among them, the variable declaration for the four parts is carried out first. The results of the four parts of Tourism revenue, Carbon footprint, Infrastructure pressure and Resident satisfaction are respectively quantified as $I(V)$, $E(V)$, $P(V)$, $R(V)$. Their respective calculation methods and descriptions are as follows:

(a) **Tourism revenue:** It is related to the average consumption level of tourists and is defined as the product of consumption level and passenger flow.

$$I(V) = C_{avg} \times \ln V \quad (1)$$

(b) **Carbon footprint:** It's related to the per capita carbon emissions of tourists.

$$E(V) = E_{avg} \times V \quad (2)$$

(c) **Infrastructure pressure:** It is linearly related to the number of tourists V .

$$P(V) = P_1 \times V + P_0 \quad (3)$$

(d) **Resident satisfaction:** It is an inverted U-shaped quadratic function curve.

$$R(V) = -f(V - V_{opt})^2 + f_0 \quad (4)$$

Among them, V_{opt} is the most suitable number of tourists for Juneau's tourism development, and f_0 is a disturbance quantity, which generally does not affect the result. By combining the above formula, the functional expression of the objective function Z is obtained. In the formula below, the weights are all non-negative (usually not 0), and the sum is 1.

$$Z = w_1 \times I(V) - w_2 \times E(V) - w_3 \times P(V) + w_4 \times R(V) \quad (5)$$

4.3 Solution Method for *STM*

4.3.1 Prerequisites and Constraints

In the previous part, we proposed a multi-objective (including double objective) optimization problem^[2], and transformed the multi-objective problem into an optimization problem for the optimal single objective Z by unifying the independent variable V . Before the specific dissolution scheme is proposed, the constraints and known terms of the above functions are listed below

(a) C_{avg} , E_{avg} , P_1 , f are variables affecting Z , V is set to 20000 people according to the number of tourists. Therefore, to ensure that the dimensions of $I(V)$, $E(V)$, $P(V)$, $R(V)$ are basically consistent, the following dimensional restrictions are made for these four variables:

- i. $C_{avg} \in [10, 10^3]$, and C_{avg} should be as large as possible, but for practical purposes, the upper bound is set to 10^3 (considering the dimension of $\ln V$ is 10 to the first power)
- ii. E_{avg} is not limited, considering that the average carbon emission and carbon footprint are affected by the environment and personal habits, so $E(V)$ should be limited, requiring $E(V) \leq E_{max} = 1500$
- iii. $P_1 \in (0, 1]$, $f \in (0, 10^{-3}]$, for the constant parameters P_0 and f_0 , assuming $P_0 = f_0 = 100$
- iv. The objective function Z should be greater than 0

(b) In formula (5), the weights $w_i, i = 1,2,3,4$ are set for the four functions. While in this part, since it revolves around Juneau, the weights are assumed to be respectively (0.4,0.3,0.2,0.1). The objective function is as follows:

$$Z = 0.4 \times I(V) - 0.3 \times E(V) - 0.2 \times P(V) + 0.1 \times R(V) \quad (6)$$

(c) The optimization model is based on the data of 2023, so the V in the model is assumed to be 20000 (according to the task), and **STM** introduces an ideal volume of tourists $V_{opt} = 15000$, more or less than this number will lead to a decline in satisfaction (V_{opt} means the number of tourists when local people are most satisfied).

4.3.2 Solution to STM: Simulated Annealing Algorithm (SAA)

In this part, based on the above constraints, in order to solve the maximum Z_{max} and C_{avg}, E_{avg}, P_1, f (case of the most optimal coefficients). **Simulated Annealing Algorithm (SAA)** is used.

This is a probabilistic optimization algorithm based on the physical annealing process, which is good at dealing with **global optimization problems**. The core idea is derived from the cooling strategy in the metal annealing process, by gradually reducing the "temperature" to find the optimal solution to the problem. This algorithm is often used to solve the traveling salesman problem (TSP), find the global optimal, function optimization problems (parameter tuning), and other aspects ^[3-5].

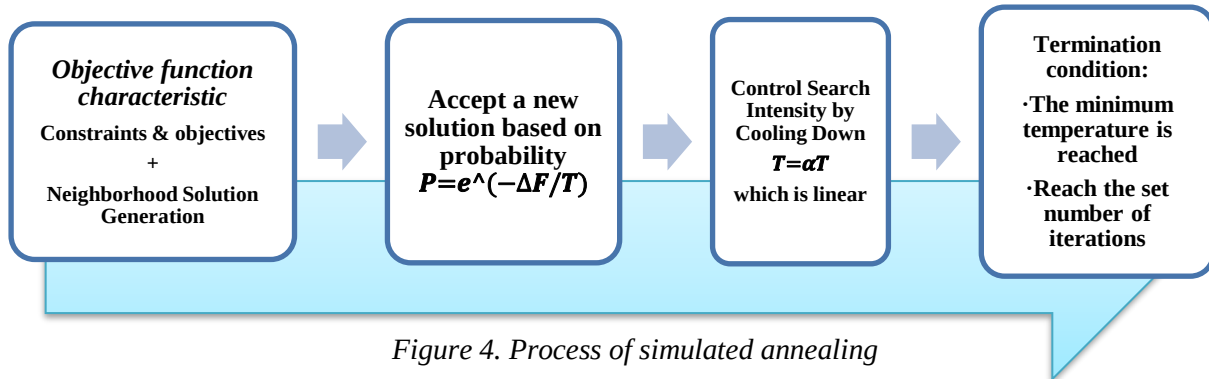


Figure 4. Process of simulated annealing

Through the above steps, **SAA** is used to solve the best Z , and its algorithm is as follows:

Algorithm of Simulated Annealing for STM

Input: $x_0, T, T_{min}, \alpha, MaxIter$

Output: x_{best}

1. Initialize $x \leftarrow x_0$
2. Compute $f(x) = Z$
3. Set $x_{best} \leftarrow x$
4. Set $f_{best} \leftarrow f(x)$
5. While $T > T_{min}$
6. For iter = 1 to $MaxIter$
7. Generate x' in the neighborhood of x
8. Compute $f(x')$
9. $\Delta f \leftarrow f(x') - f(x)$

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10.  If  $\Delta f < 0$ 
11.     $x \leftarrow x'$ 
12.  Else
13.    Accept  $x'$  with probability  $P = \exp(-\Delta F / T)$ 
14.  End If
15.  If  $f(x) < f_{best}$ 
16.     $x_{best} \leftarrow x$ 
17.     $f_{best} \leftarrow f(x)$ 
18.  End If
19. End For
20.  $T \leftarrow \alpha T$ 
21. End While
22. Return  $x_{best}$ 

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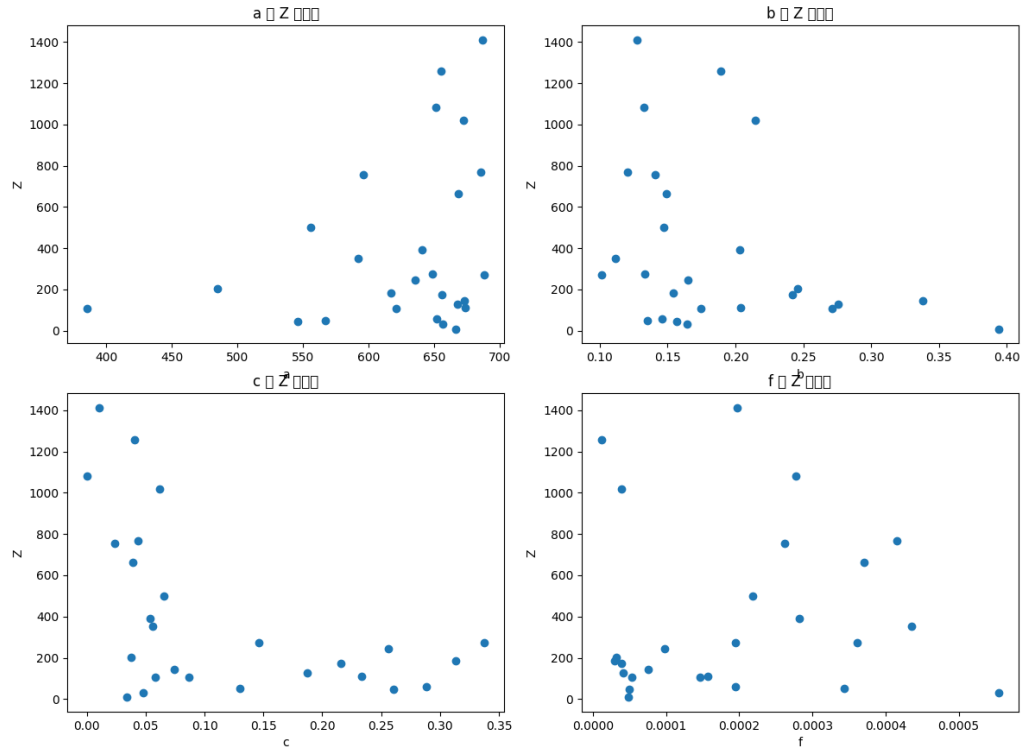
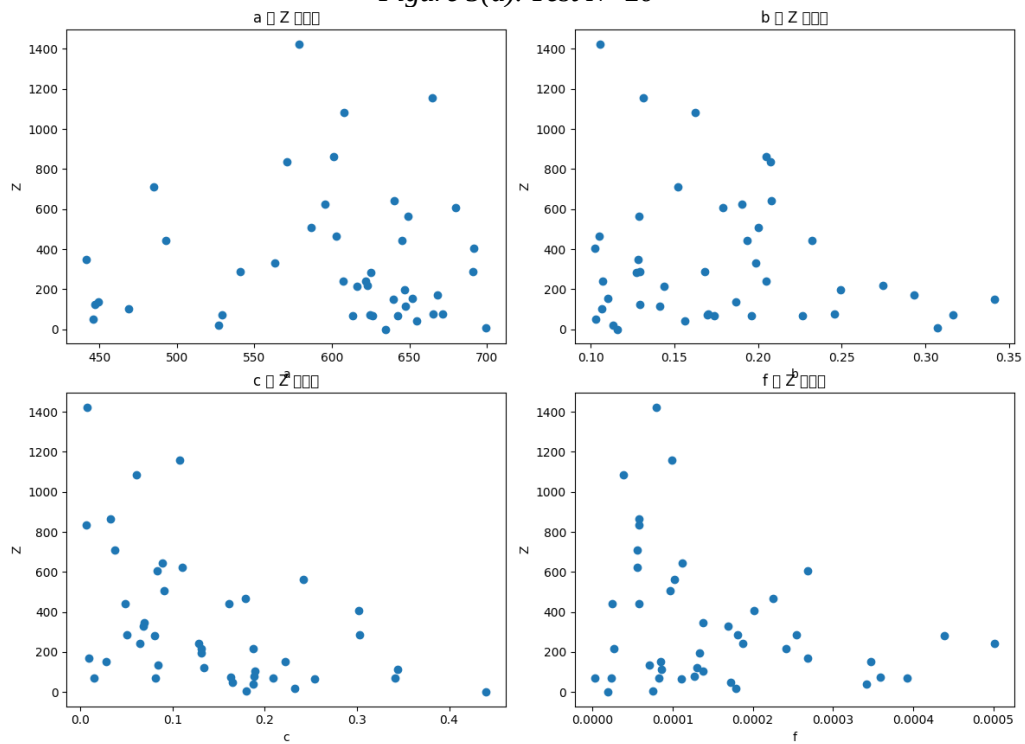
The simulated annealing algorithm seeks the **approximate global optimal solution**, rather than the absolute global optimal solution, so the simulated annealing algorithm is used to find groups of solutions, and the results are collected and plotted into scatter plots. In this paper, the parameters' mean value (MV) and variance (D) under the optimal solution of Z are obtained, and the final results are as follows:

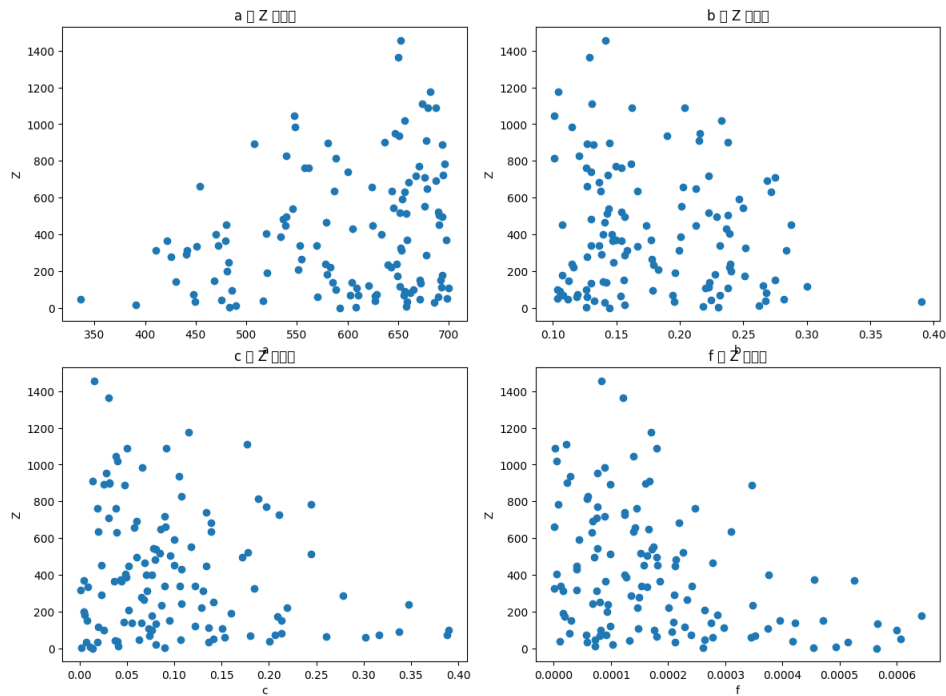
✧ SAA sets: $T = 1000, T_{min} = 1, \alpha = 0.9, MaxIter = 100$

Table 3. Coefficient of Optimization Models

Coefficient Index \ Number of Tests (N)	N=20	N=50	N=100	Average
MV- C_{avg}	624.8479	599.4152	595.9395	606.7342
D- C_{avg}	4712.6637	5197.6329	7392.0862	/
MV- E_{avg}	0.1859	0.1781	0.1795	0.181167
D- E_{avg}	0.0049	0.0038	0.0071	/
MV- P_1	0.1193	0.1419	0.1026	0.121267
D- P_1	0.01055	0.01009	0.00714	/
MV- f	0.000189	0.000155	0.000181	0.000175
D- f	2.246485	1.422288	2.237531	/
Z_{max}	397.7989	340.8236	409.0003	382.5409

For different test times, the optimal coefficient and solution obtained under each simulated annealing algorithm in the test are recorded, and the coefficient-result scatter plot is drawn as follows (in the picture, a, b, c, f correspond to coefficients C_{avg}, E_{avg}, P_1, f respectively).

Figure 5(a). Test $N=20$ Figure 5(b). Test $N=50$

Figure 5(c). Test $N=100$

According to the above data, compared with the information given in the title (375 million USD, 20,000 tourists), the per capita consumption level of tourists in Juneau still has a large room for improvement while E_{avg}, P_1 need to be further reduced to bring carbon emissions and infrastructure pressures within acceptable levels, which will lead to further reductions in hidden costs. The decrease in f value can reduce the sensitivity of residents to the dissatisfaction caused by the arrival of tourists.

5 STM Model Application

5.1 Model Case Description

The case study of Juneau was presented in the previous section, and the application of this model to other cities around the world will be explored.

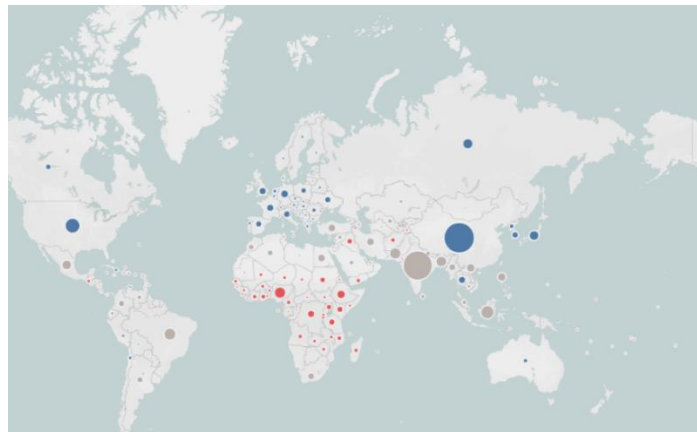


Figure 6. World Tourism Map

(data resource: <https://datacatalog.worldbank.org/dataset/world-development-indicators>)

The ideas adopted in the model are basically the same, but the development of the tourism industry in each city is different. It is because each city is different in specific aspects, so different cities need to be classified and discussed. Figure 6 shows entry and exit in the global travel industry (taking 2019 as an example), with red representing more inbound travel, blue representing more outbound travel, and grey in between. The larger the circle, the larger the data.

By summarizing the existing tourism industry, it is found that **tourism cities are mainly divided into two categories**: One is similar to Juneau, Alaska, which faces the negative impact of "tourism industry oversaturation"; The other kind of city is completely on the contrary, the small number of tourists leads to the local can not rely on tourism development, there is no high tourism revenue or perfect infrastructure, so lack of job needs, which is a negative cycle.

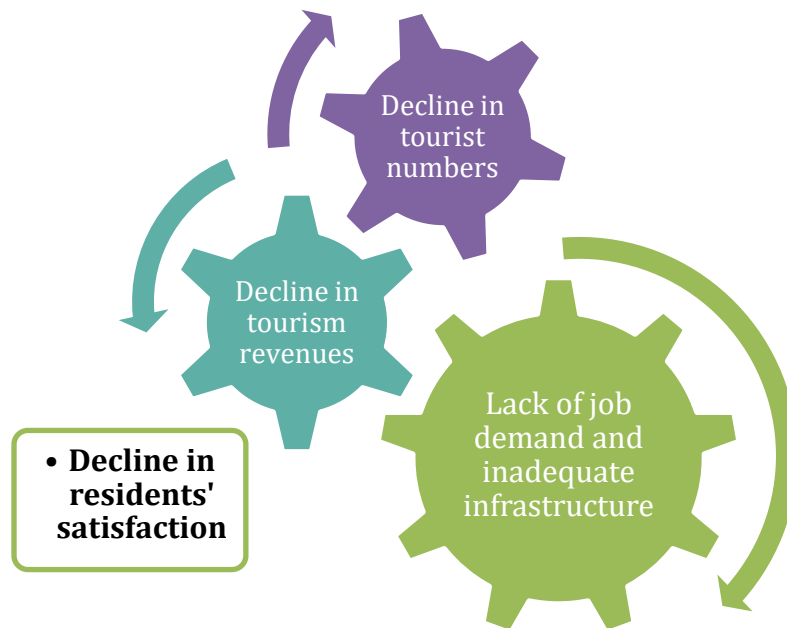


Figure 7. Problems faced by cities with under-developed tourism industry

5.2 Model Application in Barcelona & Jian Shui

Faced with the above two problems, the optimization model is basically similar to the model described in the previous part. For different cases, the weights corresponding to the indicators $I(V)$, $E(V)$, $P(V)$, $R(V)$ representing **Tourism revenue**, **Carbon footprint**, **Infrastructure pressure**, **Resident satisfaction** need to be adjusted. In the previous part, the weight set of Juneau $W = (0.4, 0.3, 0.2, 0.1)$, which needs to be adjusted here.

Before applying the model, we select cases from very different regions and cultures for two problem types of tourist cities: **Barcelona (Spain)** and **Jian Shui (Yunnan, China)**. The introduction and reasons for choosing these two examples are as follows:

Table 4. Case introduction

Case(City/Region)	Reason of choosing	Weight sets
1. Barcelona (Spain)	Spain's second-largest city, famous for its unique architectural art, rich cultural heritage, and pleasant Mediterranean scenery, attracts a	$W_1 = (0.3, 0.2, 0.3, 0.2)$

	large number of tourists every year. However, according to relevant information, the number of tourists received by Barcelona every year far <u>exceeds the number of residents</u> , and the influx of tourists has brought a series of problems to the lives of local people.	
2. Jian Shui (Yunnan, China) 	The local scenery is beautiful, characterized by the Yi ethnic group (a Chinese minority) and <u>attractive natural scenery</u> , but compared to other popular tourist cities in China, the number of tourists is small.	$W_2 = (0.5, 0.3, 0.1, 0.1)$

- a) **Case 1** $W = (0.3, 0.2, 0.3, 0.2)$ Barcelona belongs to the "oversaturated tourism industry" city, and the weight of tourism revenue should be appropriately reduced (this is different from Juneau, Barcelona does not highly rely on tourism); It does not have a high proportion of natural scenery (unlike Juneau, which has glaciers and rainforests), so the environmental pressure is not large, but the pressure on the local infrastructure and the satisfaction of the local will be affected by too many tourists.

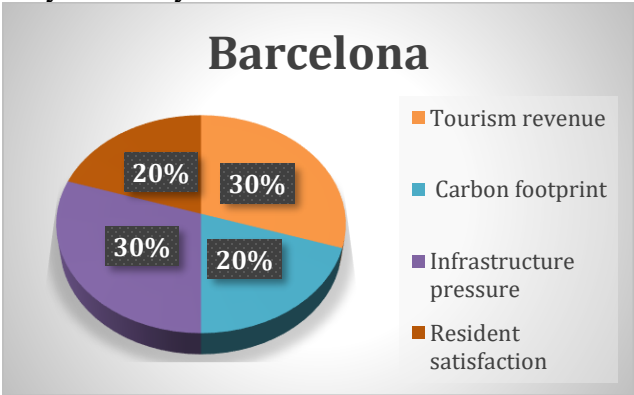
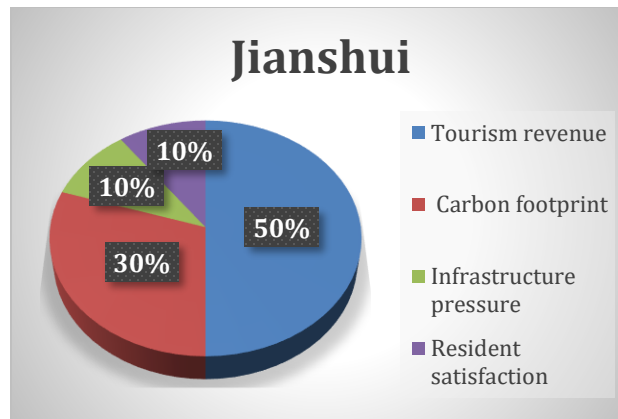


Figure 8(a). Weights of the Index of Barcelona

- b) **Case 2** $W = (0.5, 0.3, 0.1, 0.1)$ Jian Shui in Yunnan Province, China, is not popular among tourists in many local tourist cities. It is necessary to optimize and improve tourism industry income, make tourism economic income better, and protect the local ecological environment to protect the characteristic scenery. Infrastructure and the satisfaction of residents are not the most important.

Figure 8(b). Weights of the Index of *Jian Shui*

5.3 Result of Model

Based on tourist city cases in Chapter 5.2, this part will still use the **STM** model to optimize the objective function Z .

$$\begin{cases} Z_1 = 0.3 \times I(V) - 0.2 \times E(V) - 0.3 \times P(V) + 0.2 \times R(V) \\ Z_2 = 0.5 \times I(V) - 0.3 \times E(V) - 0.1 \times P(V) + 0.1 \times R(V) \end{cases} \quad (7)$$

For the above objective formula, **Simulated Annealing Algorithm** is used to solve it. City cases are tested five times each, obtaining the optimization coefficient in the following table:

Table 5(a) SAA result of case 1: *Barcelona*

TEST	No.1	No.2	No.3	No.4	No.5
C_{avg}	451.5881	456.1657	497.136	163.5263	918.0977
E_{avg}	1.7046	3.9678	1.5892	4.5721	1.2593
P_1	0.005386	0.002039	0.006754	0.000365	0.001832
f	0.000682	0.000678	0.00047	0.000296	0.000271
Z	7.2907	7.2969	7.6634	8.1252	8.2128

Table 5(b) SAA result of case 2: *Jian Shui*

TEST	No.1	No.2	No.3	No.4	No.5
C_{avg}	297.052	309.2254	68.521	66.2924	116.1655
E_{avg}	0.43592	2.0504	2.0463	3.212	0.38806
P_1	0.005683	0.009132	0.005524	0.003127	0.004452
f	0.000976	0.000919	0.000702	0.00074	0.000676
Z	6.9322	6.992	7.2616	7.2092	7.2987

After the data is processed, the minimum value of the 10 test results of the unified parameter is defined as 2, and the maximum value is 9. The remaining data is projected into the range of 2 to 9. In this way, the order of magnitude of data is erased and the data is unified to facilitate clear observation of data changes and discovery rules. The results are shown in the figure below.

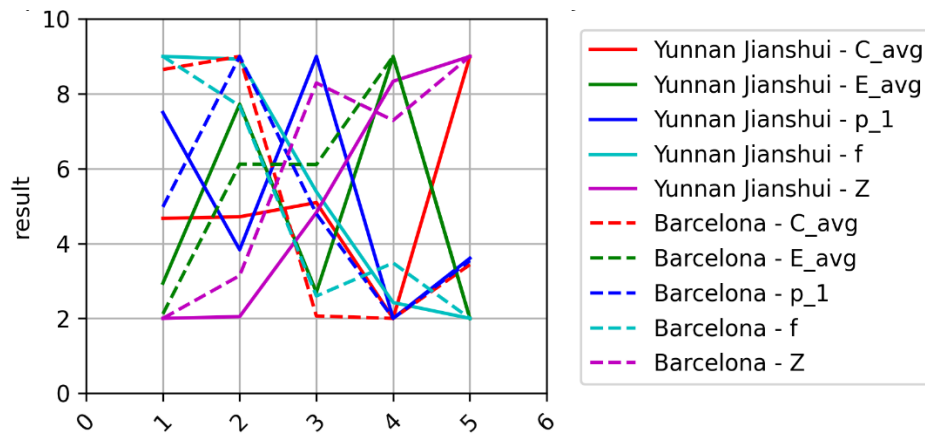


Figure 9. Comparison between Jian Shui and Barcelona

Through the above data, it is found that Barcelona and Jian Shui have obvious differences in optimization priorities due to different tourism development conditions:

- Barcelona has a huge number of tourists (the Airport has more than 50 million passengers in the first eight months of 2024), and the local government is faced with problems such as environmental pollution and infrastructure bearing pressure caused by excessive tourists. The optimization focuses on the **optimization of the environment and infrastructure** as well as the **adjustment of the tourism industry structure**.
- There are relatively few tourists in Jian Shui, and the tourism development potential is not fully tapped. The focus is on the **development and promotion of tourism resources** and the improvement of **tourism supporting facilities**, to stimulate the income of tourism.



Figure 10. How to optimize Jian Shui Ancient City and Barcelona

Barcelona, Spain

- In terms of environmental and infrastructure optimization, it is necessary to strengthen pollution control. Excessive gathering of tourists leads to an increase in garbage, causing pollution to beaches, streets and other environments. Increase the frequency of garbage disposal, improve garbage sorting facilities, and improve the efficiency of garbage disposal.
- Barcelona's infrastructure bears a lot of pressure, traffic congestion around popular attractions, and water and electricity supply is tight. It is necessary to widen roads, optimize public transport lines, increase investment in water and electricity supply facilities, and improve their carrying capacity.

- iii. Adjust the structure of the tourism industry, the flow and distribution of tourists should be regulated. Local governments can control the number of tourists by raising ticket prices for popular attractions and implementing time-sharing visits. Develop new tourist attractions to divert tourists away from popular areas and relieve the pressure on scenic spots.
- iv. Improve the quality of tourism, guide merchants to improve the quality of goods and services, standardize the market order, reduce the sale of shoddy souvenirs, combat the behavior of rip-off customers, and maintain the image of urban tourism.

Jian Shui, Yunnan, China

- i. In terms of tourism resources development and publicity, the characteristic tourism resources should be deeply explored. The preservation and development of local ancient buildings, and the addition of interactive experience projects, so that tourists can deeply feel the historical and cultural charm; Hold a food culture festival to promote Jian Shui tofu, steamed chicken, and other special foods to attract tourists.
- ii. Strengthen publicity and marketing. Make use of online platforms and social media for online publicity, and produce exquisite short videos and graphic introductions to show the beauty, cuisine, and culture of Jian Shui Ancient City; Participate in international tourism exhibitions to raise awareness.
- iii. In the improvement of tourism supporting facilities, improve the transportation facilities. Increase flights to Jian Shui Scenic area, and high-speed rail shuttle buses to facilitate the arrival of tourists; Increase the public transport lines and vehicles in the city, and set up tourist special lines to facilitate tourists to shuttle between scenic spots. Improve accommodation, catering and other supporting facilities.
- iv. Encourage the construction of hotels and homestays of different grades to meet the needs of various tourists; Set up special food blocks, standardize the standards of catering services, and improve the accommodation experience of tourists.

When analyzing the development of different tourist cities and giving suggestions for them, **we should consider the fundamental problems** of the cities in combination with reality. For example, in the ancient city of Jian Shui in the case, its development level is not as good as that of large cities, its infrastructure and population size are small, and the city itself relies on tourism development, so it needs to strengthen the protection of scenic spots. Both Barcelona and Juneau are facing the negative effects caused by too many tourists. The difference is that Barcelona relies on humanistic architecture and culture to attract tourists, which does not depend on tourism and development, while Juno relies on tourism to develop, which is also different.

6 Sensitivity Analysis of *STM*

In the fourth part of the article, the ***STM*** model is explained and involves the corresponding function of 4 development evaluation objectives and the final objective. In this section, the effect of changes in coefficients on the final target Z_{max} will be discussed. In the case of adjustment of model parameters and functions, observe whether the final goal of the model changes and explore the coefficient that has a greater impact on the goal (the corresponding evaluation goal behind it) and the law between them.

In the Tourism industry revenue, **Tourism revenue** will be discussed, because its corresponding function $I(V)$ is set as $I(V) = C_{avg} \times \ln V$ in the previous model, indicating that the relationship between it and the passenger flow is non-linear (to be precise, the curve of

increasing speed becomes slower). So what happens when it becomes $I(V)' = C_{avg} \times V$, the same linear relationship as **Carbon emission** ($E(V)$) and **Infrastructure pressure** ($P(V)$)?

At the same time, **Resident satisfaction** is also a variable worth studying, because unlike other parts showing a monotonous trend of change, it's a law of **first increasing and then decreasing** with the passenger flow V . This is also because when the number of tourists is too large or too small, it will lead to the dissatisfaction of local residents, so it is necessary to set an optimal number of tourists V_{opt} (specific details have been described above). In its function, the coefficient f ($f > 0$) represents the opening size of the inverted U-type quadratic function, and there is a square term, which has a great influence on the whole and is worth discussing.

Next, we will discuss the situation when the **Tourism revenue** function model changes or the f value changes (Figure 11). When $I(V)$ is logarithmic or linear, the influence of different f values on the optimal passenger volume V and Z_{max} . (Note that the optimal passenger volume here is not the ideal passenger volume V_{opt})

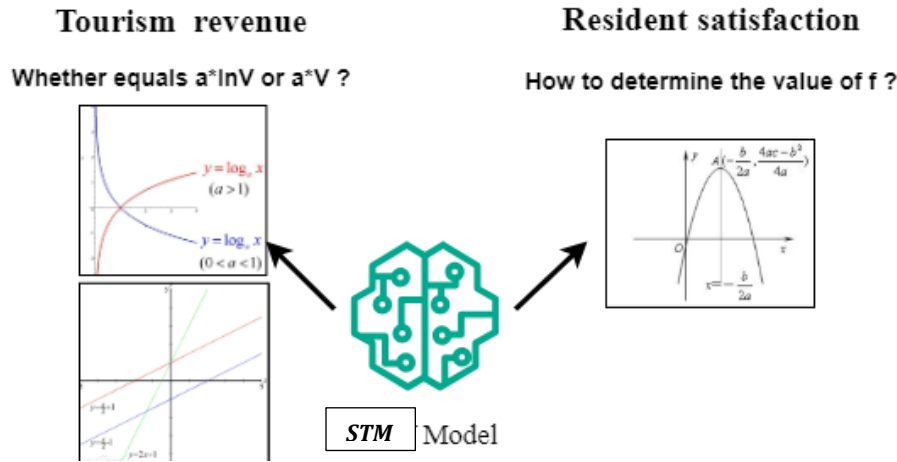


Figure 11. Different conditions of model parameter changes

6.1 Case A

Leave the weight Settings unchanged. When $I(V)$ is a **logarithmic function**, the final optimization function is consistent with the formula in the fourth part, which is still:

$$Z = 0.4 \times C_{avg} \times \ln V - 0.3 \times E(V) - 0.2 \times P(V) + 0.1 \times R(V) \quad (7)$$

Where, the Resident satisfaction function $R(V)$ is expressed as:

$$R(V) = -f(V - V_{opt})^2 + f_0 \quad (8)$$

Assume that $C_{avg} = 187.5$, $E_{avg} = 0.1$, $P_1 = 0.01$, $P_0 = 100$

Next, the opening size of the quadratic function is discussed. f reflects the psychological acceptance degree of residents for the gap between the number of tourists and the ideal number of tourists, which can also be understood as the sensitivity of residents to the number of tourists. Set value of $f_A = (0.0001, 0.001, 0.01, 0.1, 1, 2, 5, 10, 20, 50)$, $f_0 = 0$. As above, the simulated annealing method is used to find the optimal tourist volume V_{best} and a corresponding return Z_{max} . The specific data obtained by calculation are shown in the table below.

Table 6(a). Results of V_{best} & Z_{max} in **Case A**

f	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	2	5	10	20	50
V_{best}	1367	1486	1498	1499	1499	1499	1499	1499	1499	1499
	4.238	5.226	6.502	8.65	9.865	9.932	9.973	9.986	9.993	9.997
	3	7	3			5		6	2	3
Z_{max}	239.0	223.0	221.3	221.2	221.1	221.1	221.1	221.1	221.1	221.1
	931	049	676	036	872	863	857	856	855	854

6.2 Case B

Again, leave the weight Settings unchanged. When $I(V)$ is a **linear function**, the final optimization function is consistent with the formula in the fourth part, which is still:

$$Z = 0.4 \times C_{avg} \times \ln V - 0.3 \times E(V) - 0.2 \times P(V) + 0.1 \times R(V) \quad (9)$$

Where the $R(V)$ function remains unchanged, and the set of values for f_B is the same as that for f_A . The **Simulated Annealing** method is also used to obtain the results:

Table 6(b). Results of V_{best} & Z_{max} in **Case B**

f	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	2	5	10	20	50
V_{best}	1999	1999	1999	1874	1537	1518	1507	1503	1501	1500
	9.999	9.999	9.999	8.4	4.84	7.42	4.968	7.484	8.742	7.496
	4	4	4							8
Z_{max}	1499	1496	1474	1265	1138	1131	1127	1125	1125	1124
	089.9	839.9	339.9	005.0	550.5	525.2	310.1	905.0	202.5	781.0
	55	56	61	26	03	51	01	5	25	1

6.3 Case A VS. Case B

Through the above calculation, when $I(V)$ is a logarithmic function (Case A log-type) and a linear function (Case B linear-type), as the value of f gradually increases (order of magnitude increases), the changes of V_{best} (blue line) and Z_{max} (red line) are regular:

- As the value of f increases, V_{best} in Case A increases and V_{best} in Case B decreases;
- When the value of f increases, both converge to V_{opt} ;
- In Case A, V_{best} converges faster and earlier to V_{opt} . In Case B, V_{best} converges more slowly and converges to V_{opt} later.
- The final result Z_{max} of Case A and Case B has a large difference in order of magnitude, but both tend to converge to a certain number. Similarly, Case A converges faster than Case B.

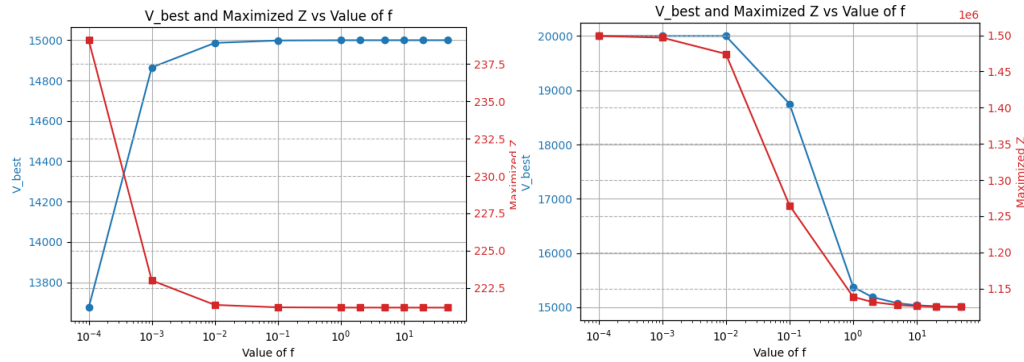


Figure 12. V_{best} & Z_{max} of Case A (left) and Case B (right)

In addition, the article observes the case of the **Resident satisfaction** function $R(V)$ (Figure 13(a, b)). In either case, when the order of f is small (f is less than 10 to the power of 0), the size of the opening does not change significantly, and when the coefficient f order of magnitude reaches a certain level, it will change significantly.

It is not difficult to find that only after the change of f reaches a certain order of magnitude, the impact on the result will be greater. The change in the tourism industry income model will have a great impact on the overall result. As a tourist city, it is more important to seize more opportunities to attract tourists, especially in Case A (logarithmic function). As long as the number of tourists V is within a certain reasonable range (do not exceed V_{opt} too much), the economic benefits brought by attracting tourists will offset the hidden costs brought by infrastructure and residents' satisfaction.

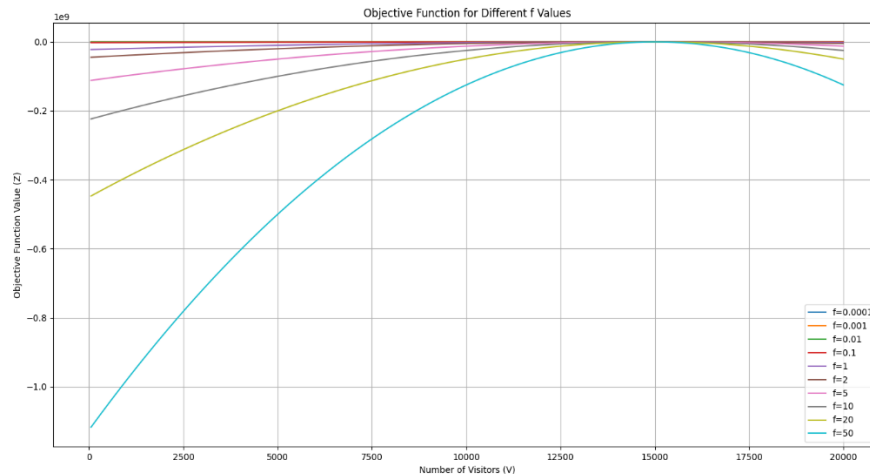


Figure 13(a). $R(V)$ in Case A (log-type)

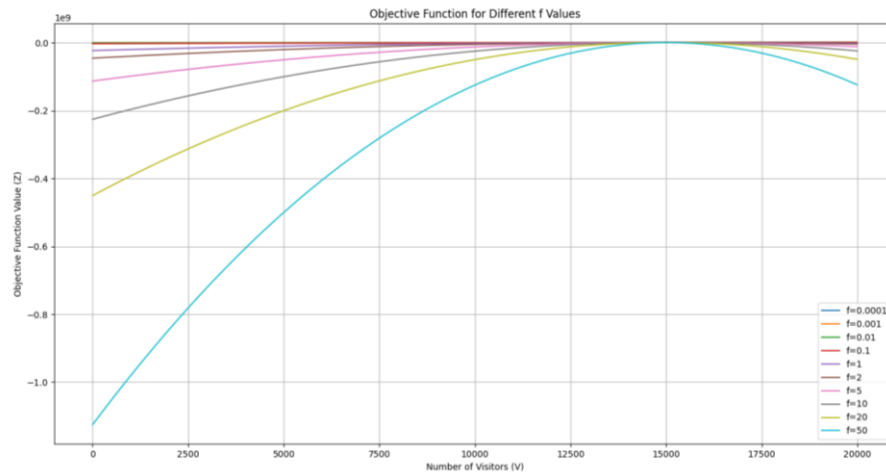


Figure 13(b). $R(V)$ in **Case B** (linear-type)

7 Prediction for Juneau

From the above analysis, we can see that the economic, social, and environmental aspects of the tourism industry are closely related to the volume of tourists. To better grasp the future development direction of the Juneau tourism industry, it is very important to predict the number of future tourist populations. Based on the number of ship visitors to Juneau over the past 10 years in the appendix (since 98% of all tourist visitors to Juneau take cruises, the number of cruise visitors can be roughly taken as the number of visitors), the number of future visitors can be predicted.

In this paper, the **ARIMA time series model** is used to predict the number of future tourist visits. This is a prediction method applicable to data sets with no rules to follow in the short and medium term^[6-7], in which there are three important variables p, d, q . Among them, p and q respectively represent the order of autoregressive term (AR) and moving average (MA), while d is the difference order, which represents the number of deviations between data.

Through the difference between the disordered data, find the relationship between the original data through the difference, usually $d = 0, 1, 2$. In this paper, $d = 0, 1, 2$ are calculated respectively, and the data for the next three years (2024, 2025, 2026) are predicted. The data sets for the past 10 years (2014-2023) are as follows:

$$V(t) = (961000, 983000, 1015000, 1072000, 1151000, 1306000, 0, 117000, 1167000, 1670000)$$

Table 7. Prediction of **Tourist Volume** by ARIMA

$d \setminus V(t)$	$t = 2024$	$t = 2025$	$t = 2026$
0	1410639 ↓	486552 ↓ ↓	721272 ↑
1	1260129 ↓ ↓	1026659 ↓	635584 ↓
2	1742053 ↑	1097802 ↓	1475914 ↑

As can be seen from the table, due to the impact of COVID-19 in 2020 and 2021, the subsequent forecast data **shows signs of decline**. In combination with the actual situation, the

prediction results when $d = 2$ are relatively valuable (In Figure 14, where the red line is the predicted value and the blue line is the historical value).

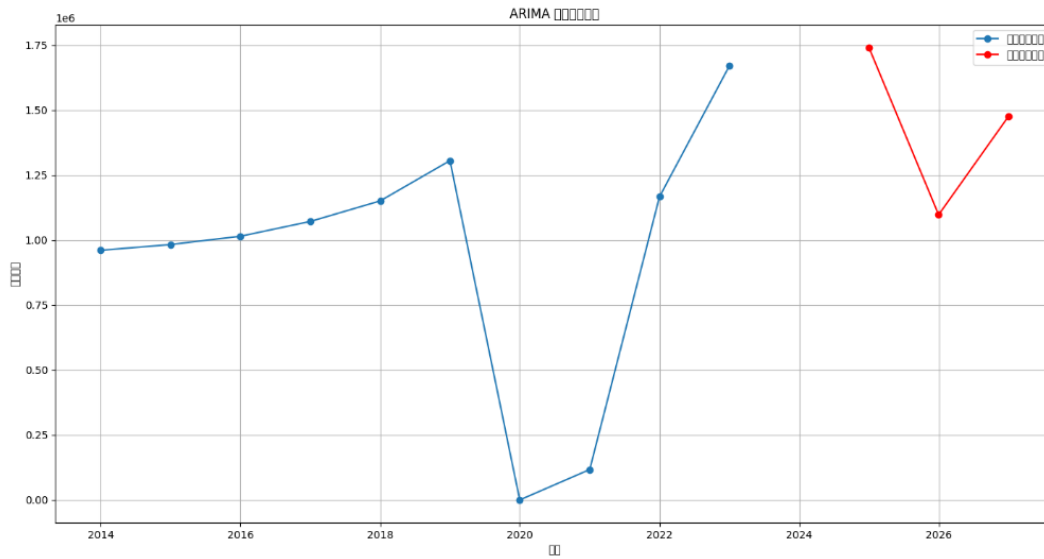


Figure 14. Prediction by ARIMA and the growing trend of volume

8 Conclusion

With the rapid development of the tourism industry, we analyze the current status of the tourism industry for Juneau, Alaska, and put forward optimization plans and give suggestions for the future development of the local tourism industry. A sustainable tourism industry optimization model (**STM**) is proposed, and some mathematical algorithms can be used to optimize its internal parameters, and then relative conclusions are drawn.

8.1 Strengths

- The optimization model used is a classic multi-objective planning design idea. The target is the total revenue of the tourism industry. The maximum total revenue can be achieved by optimizing the parameters.
- To solve the multi-objective planning problem, we transform the multi-objective problem into a single-objective problem, and the local tourism income is mainly considered. The hidden cost factor is deducted from the net income as a penalty.
- There are many solutions to the optimization problem (particle swarm optimization algorithm, genetic algorithm GA). Here, simulated annealing algorithm (**SAA**) is selected to solve the optimization parameters
- Obtain the **main independent variable**: volume of tourist passengers V , and select a suitable prediction model to forecast it, select the forecast data close to reality, and give suggestions.
- The weights of different targets (evaluation indicators) are also set, and **different weight** coefficients are set under different circumstances, which makes the model flexible.

8.2 Weaknesses

- There is a lack of further discussion of the model, such as how to set the coefficient of carbon emissions, and whether there are other ways to construct the function of infrastructure stress.
- The model needs more data support. More data can be collected to make the model more accurate.

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